

# QUANTITATIVE ECONOMICS - SPRING 2021

Answer any FIVE questions.

1. (a) Consider the regression model

$$Y_i = \beta X_i + u_i \quad i = 1, \dots, n$$

where  $u_i$  and  $X_i$  satisfy the standard assumptions. Let  $\bar{\beta} = \frac{\bar{Y}}{\bar{X}}$ , where  $\bar{Y}$  and  $\bar{X}$  are the sample means of  $Y_i$  and  $X_i$ , respectively.

- i. Show that  $\bar{\beta}$  is a linear function of  $Y_1, Y_2, \dots, Y_n$ .
  - ii. Show that  $\bar{\beta}$  is conditionally unbiased.
  - iii. Derive the variance of  $\bar{\beta}$ .
- (b) Consider the regression model  $Y_i = \beta X_i + \varepsilon_i$ . Suppose that a researcher mistakenly applies the least squares method to  $X_i = \delta Y_i + u_i$  and then uses  $\frac{1}{\delta}$  as an estimator for  $\beta$ . Find the bias of  $\frac{1}{\delta}$ .
2. Consider the regression model

$$y_i = \alpha + \beta x_i + u_i, \quad i = 1, 2, \dots, n$$

- (a) Suppose we assume that  $\alpha = 0$  and hence  $y_i = \beta x_i + u_i$
- i. Write down the moment condition(s) used to derive the method of moments estimator for  $\beta$ .
  - ii. Write down the least-squares objective function to derive the OLS estimator for  $\beta$ .
  - iii. Use the equation from ii. to derive the estimator for  $\beta$ .
- (b) Suppose we assume that  $\beta = 0$  and hence  $y_i = \alpha + u_i$
- i. Write down the moment condition(s) used to derive the method of moments estimator for  $\alpha$ .
  - ii. Write down the least-squares objective function to derive the OLS estimator for  $\alpha$ .
  - iii. Use the equation from ii. to derive the estimator for  $\alpha$ .
3. Suppose you collect data from a survey on wages, education, experience, and gender. In addition, you ask for information about marijuana usage. The original question is: "On how many separate occasions last month did you smoke marijuana?"
- (a) Write an equation that would allow you to estimate the effects of marijuana usage on wage, while controlling for other factors. You should be able to make statements such as, "Smoking marijuana five more times per month is estimated to change wage by x%."
  - (b) Write a model that would allow you to test whether drug usage has different effects on wages for men and women. How would you test that there are no differences in the effects of drug usage for men and women?
  - (c) Suppose you think it is better to measure marijuana usage by putting people into one of four categories: nonuser, light user (1 to 5 times per month), moderate user (6 to 10 times per month), and heavy user (more than 10 times per month). Now, write a model that allows you to estimate the effects of marijuana usage on wage.
  - (d) Using the model in part (c), explain in detail how to test the null hypothesis that marijuana usage has no effect on wage. Be very specific and include a careful listing of degrees of freedom.
4. Consider the following data generating process

$$Y_t = c + \phi_2 Y_{t-2} + \varepsilon_t$$

where  $\varepsilon_t$  is a white noise process. Assuming  $0 < \phi_2 < 1$ , answer the following:

- (a) Derive the expected value of this process.
- (b) Derive the variance of this process.

- (c) Derive the covariance of this process for  $j = 1, 2, 3$ .
- (d) Derive the autocorrelation function of this process for  $j = 1, 2, 3$ .
5. In a fixed effects model, we can consider several ways to control for the individual effects which may be correlated with the regressors. Three cases being the within estimator, the least-squares dummy variable estimator (LSDV) and the first difference estimator. Consider the three methods, show that
- (a) The within estimator and the LSDV estimator are equivalent.
- (b) When  $T = 2$ , the within estimator and the first difference estimator are equivalent.
6. Suppose we have data for U.S. interest rate data over the period 1960:Q1 to 2008:Q1 and we are interested in estimating a quarterly model of spread between a long-term and a short-term interest rate. Specifically, the interest rate spread ( $s$ ) can be formed as the difference between the interest rate on a 10-year U.S. government bonds ( $r10$ ) and the rate on a three-month treasury bills ( $Tbill$ ) as

$$s_t = r10_t - Tbill_t$$

Suppose the interest rate spread in the fourth quarter of 2007 is 0.75, the interest rate spread in the first quarter of 2008 is 1.51 and the interest rate spread in the second quarter of 2008 is 1.61. Use the EViews output to answer the following questions:

| <b>Dependent Variable: S</b><br><b>Method: Least Squares</b><br><b>Date: 12/30/09 Time: 10:28</b><br><b>Sample (adjusted): 1960Q3 2008Q1</b><br><b>Included observations: 191 after adjustments</b><br><b>Convergence achieved after 3 iterations</b> |             |                       |             |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------|-------------|----------|
| Variable                                                                                                                                                                                                                                              | Coefficient | Std. Error            | t-Statistic | Prob.    |
| C                                                                                                                                                                                                                                                     | 1.387620    | 0.285227              | 4.864963    | 0.0000   |
| AR(1)                                                                                                                                                                                                                                                 | 1.109152    | 0.070829              | 15.65953    | 0.0000   |
| AR(2)                                                                                                                                                                                                                                                 | -0.245386   | 0.070792              | -3.466283   | 0.0007   |
| R-squared                                                                                                                                                                                                                                             | 0.805374    | Mean dependent var    |             | 1.378220 |
| Adjusted R-squared                                                                                                                                                                                                                                    | 0.803304    | S.D. dependent var    |             | 1.210806 |
| S.E. of regression                                                                                                                                                                                                                                    | 0.536998    | Akaike info criterion |             | 1.609936 |
| Sum squared resid                                                                                                                                                                                                                                     | 54.21291    | Schwarz criterion     |             | 1.661019 |
| Log likelihood                                                                                                                                                                                                                                        | -150.7489   | F-statistic           |             | 388.9784 |
| Durbin-Watson stat                                                                                                                                                                                                                                    | 1.960866    | Prob(F-statistic)     |             | 0.000000 |
| Inverted AR Roots                                                                                                                                                                                                                                     | .80         | .31                   |             |          |

- (a) Write the estimated equation given in the Eviews output.
- (b) Obtain the one-step ahead forecast.
- (c) Obtain the one-step ahead forecast error.
- (d) Obtain the two-step ahead forecast.
- (e) Obtain the three-step ahead forecast.
7. A variable  $Q$  is determined by the model

$$Q = \beta_1 + \beta_2 X + v$$

where  $X$  is a variable and  $v$  is a disturbance term that satisfies the regression model assumptions. The dependent variable is subject to measurement error and is measured as  $Y$  where

$$Y = Q + r$$

and  $r$  is the measurement error, distributed independently of  $v$ . Describe analytically the consequences of using OLS to fit this model if:

- (a) The expected value of  $r$  is not equal to zero (but  $r$  is distributed independently of  $Q$ ).
- (b)  $r$  is not distributed independently of  $Q$  (but its expected value is zero).
8. Suppose I am thinking about figuring out how much my rent will be given my income. I have gathered data on Washington, DC apartment rents (\$/Year) and the incomes of the people who live in those apartments (\$/Year), as well as the square footage of the apartment and whether it has an unobstructed view (Dummy Variable). Two regressions have been run and the results are reported in the table below.

| Regressor/Dependent | Rent                | $\hat{u}^2$                |
|---------------------|---------------------|----------------------------|
|                     | Model 1             | Model 2                    |
| Intercept           | 5455.48<br>(602.77) | -361594.30<br>(5330126.00) |
| Income              | 0.0635<br>(0.0143)  | 305.6033<br>(127.2458)     |
| Income <sup>2</sup> | -0.0002<br>(0.0084) | 40.3662<br>(19.3689)       |
| SquareFoot          | 0.0234<br>(0.0074)  | -59633.25<br>(28966.258)   |
| View                | 1000.26<br>(206.93) | -26988.3698<br>(30599.358) |
| Observations        | 108                 | 108                        |
| R <sup>2</sup>      | 0.1555              | 0.0516                     |

- (a) Interpret the impact of income on yearly rent?
- (b) Conduct a test of no regression in model (1) at the 5% level. Be sure to write the null and alternative hypothesis correctly.
- (c) Explain why the error variance of the model run in model (1) may be heteroscedastic with regard to income.
- (d) Consider model (2). Here the dependent variable is the square of the residuals from model (1). What type of test can be implemented from model (2)? Perform this test at the 10% level.
- (e) Does the test in (e) specifically test if the error variances are heteroscedastic with regards to income? Why or why not?

**PLEASE SEE NEXT PAGE**

**Following questions are for those students who took ECOG 216: Adv. Topics in**

**Econometrics with Dr. Amini:**

9. Let the variable Productivity denote a factory worker  $i$ 's total output per hour, and let the variable Training denote the hours of additional training for that worker. We want to investigate the effect of additional training on workers' productivity. Let's start with a simple model:

$$PRODUCTIVITY_i = \beta_0 + \beta_1 TRAINING_i + U_i$$

where  $U_i$  is the unobserved error.

- (a) What kinds of factors are contained in  $U$ ? Are these likely to be correlated with level of training?
- (b) Will a simple regression analysis uncover the ceteris paribus effect of training on productivity? Explain.
- (c) In order to address the causal link between the two variables, design a randomized experiment and write down a regression model based on your proposed experiment.
- (d) In your designed randomized experiment model, show that the observed difference in the dependent variable average is equal to the treatment effect and the potential selection bias.
- (e) Explain how randomized experiment can solve the selection bias problem.

10. We want to estimate the causal impact of the variable  $S_i$  on the Variable  $Y_i$ , using an instrument variable  $Z_i$  in the following model with covariates:

$$Y_i = \alpha'X_i + \rho S_i + \eta_i,$$

- a) Write down the first stage and the reduced form equations for the IV model.
- b) Show that the IV estimator in your model can be written as a function of two coefficients from the first stage and reduced form equations.
- c) For a Two-Stage Least Squares model, derive the reduced form equation using the first stage equation and the above the causal relation of interest.
- d) Explain the exclusion restriction and the strong first stage assumptions for a using IVs.

11. In regression discontinuity models:

- What is the difference between Sharp and Fuzzy RD strategy?
- For the Sharp RD model, derive a regression formula that includes the discontinuity variable. Explain the causal implication of the discontinuity coefficient.
- For the fuzzy RD, derive a regression formula that includes the discontinuity variable. Also, you need to write both the first stage and the reduced form models. Explain the meaning of equations.
- Lee's (2008) study of the effect of party incumbency on re-election probabilities using the sharp RD design with the treatment status  $D_i$

$$D_i = \begin{cases} 1 & \text{if } x_i \geq x_0 \\ 0 & \text{if } x_i < x_0 \end{cases}$$

The diagram below illustrated the paper's Sharp RD design. Explain what is  $D_i$  and What is  $X_i$  in Lee's paper. Also explain the result you see in the diagram. (5 points)

